lower concentrations of contaminants more effectively. The strategies for bioremediation in soil and water can be as follows:

- Use of indigenous microbial population as indicator species for bioremediation process.
- Bioremediation with the addition of adapted or designed microbial inoculants.
- Use of plants for bioremediation - green technology.

Some examples of bioremediation technologies are:

- **Phytoremediation** - use of plants to bring about remediation of environmental pollutants.
- **Mycoremediation** - use of fungi to bring about remediation of environmental pollutants.
- **Bioventing** a process that increases the oxygen or air flow to accelerate the degradation of environmental pollutants.
- **Bioleaching** use of microorganisms in solution to recover metal pollutants from contaminated sites.
- **Bioaugmentation** a addition of selected microbes to speed up degradation process.
- **Composting** process by which the solid waste is composted by the use of microbes into manure which acts as a nutrient for plant growth.
- **Rhizofiltration** uptake of metals or degradation of organic compounds by rhizosphere microorganisms.
- **Rhizostimulation** stimulation of plant growth by the rhizosphere by providing better growth condition or reduction in toxic materials.

Limitations

- Only biodegradable contaminants can be transformed using bioremediation processes.
- Bioremediation processes must be specifically made in accordance to the conditions at the contaminated site.
- Small-scale tests on a pilot scale must be performed before carrying out the procedure at the contaminated site.
- The use of genetic engineering technology to create genetically modified microorganism or a consortium of microbes for bioremediation process has great potential.

4.8.12 Biofuel: Algal Biofuel

Algal fuel, also known as algal biofuel, or algal oil is an alternative to liquid fossil fuels, the petroleum products. This is also used as a source of energy-rich oils. Also, algal fuels are an alternative to commonly known biofuel sources obtained from corn and sugarcane. The energy crisis and the world food crisis have initiated interest in algal culture (farming algae) for making biodiesel and other biofuels on lands unsuitable for agriculture. *Botryococcus braunii* is normally used to produce algal biofuel.

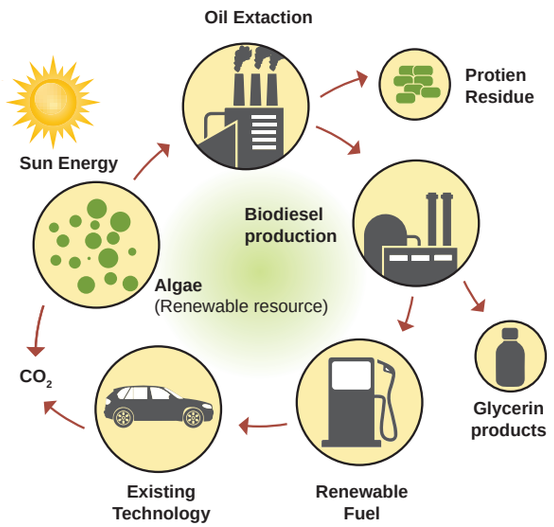


Figure 4.31: Algal Biofuel

Biological hydrogen production by algae

The biological hydrogen production with algae is a method of photo biological water splitting. In normal photosynthesis the alga, *Chlamydomonas reinhardtii* releases oxygen. When it is deprived of sulfur, it switches to the production of hydrogen during photosynthesis and the electrons are transported to ferredoxins.

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